

Shiyu (Rayne) Xu

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EDUCATION

University of Michigan - Ann arbor

Aug 2020 – May 2022 | Ann arbor, MI

Master of Science in Information (Human Computer Interaction)

Cumulative GPA: 4.0/4.0

Advisor: Prof. Steve Oney

University of Wisconsin Madison

Sep 2016 – May 2020 | Madison, WI

B.S in Computer Science, Certificate in Graphic Design & Mathematics

GPA: 3.7/4.0

PUBLICATIONS

MIND: Empowering Mental Health Clinicians with Multimodal Data Insights through a Narrative Dashboard

Shiyu Xu*, Ruishi Zou*...

Under Review

How Pairing by Code Similarity Influences Discussions in Peer Learning

Shiyu Xu, Ashley Ge Zhang, Steve Oney

Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems

“It would probably turn into a social faux-pas”: Users’ and Bystanders’ Preferences of Privacy Awareness Mechanisms in Smart Homes.

Parth Kirankumar Thakkar, Shijing He, Shiyu Xu, Danny Yuxing Huang, Yaxing Yao

Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems

RESEARCH EXPERIENCE

Research on Mental Health

April 2025 - Sep 2025

Columbia University

Advisor: Xuhai Xu

MIND (Multimodal Data Integrated Narrative Dashboard) is a clinician-facing system designed to support mental health care by transforming complex multimodal patient data, such as clinical notes, survey scores, transcripts, and passive sensing data into coherent, data-driven narratives dashboard.

- Co-designed MIND with five mental health clinicians through an iterative process.
- Developed the MIND interface using React, supported by a hybrid backend pipeline that integrates large language models (LLMs) with rule-based analytics to generate and display clinically relevant insights.
- Quantitatively and qualitatively analyzed user study results with 16 clinicians, demonstrating that MIND significantly improved insight discovery ($p < .001$), decision-making ($p = .004$), data integration ($p < .001$), and presentation efficiency ($p = .006$) compared to traditional dashboards in simulated patient sessions.

Research on Peer programming in Computer Science Online Education

Sept 2021 - May 2022

University of Michigan - Ann Arbor

Advisor: Steve Oney

- Evaluated the effectiveness of grouping students by solution similarity in programming courses and its impact on their discussions and future coding diversity.
- Conducted one-hour user studies with 15 participants, collected and analyzed quantitative survey data and qualitative observations, using Excel and matrix analysis to evaluate participants' coding skills, interactions, and problem-solving processes.

Research on Smart Home Privacy

May 2020 - Oct 2020

University of Maryland

Advisor: Yaxing Yao

- Evaluated preferences and effectiveness of privacy-related notifications in smart homes, identifying the pros and cons of four privacy awareness mechanisms and key design considerations for future improvements.
- Conducted a survey with 136 users and 123 bystanders, analyzing the results using matrix analysis and content analysis to evaluate privacy-related notifications in smart homes.

PROFESSIONAL EXPERIENCE

Moderna, Inc

May 2023 – Now

Software Engineer

- Contributed to the front-end development of Moderna Direct's e-commerce site (facilitating the purchase of Moderna's vaccine products). Utilizing tools such as React, JavaScript, Redux, and Material UI.
- Collaborated cross-functionally with UX designers and engineers through extensive meetings to bridge communication gaps.
- Translated Figma UI designs into functional code using the Material-UI component library, ensuring WCAG-compliant accessibility standards; integrated frontend with backend services using Redux for state management and API interactions.

Zoom Video Communications, Inc

Jul 2022 – May 2023

Front-end Engineer

- Implemented essential features for Zoom's Chat platform, utilizing ReactJS as the primary framework and employing Typescript, SASS, and HTML5 as coding languages, with Mobx and RxJS for state management.
- Analyzed user behavior and identified 5+ pain points, creating user workflows and proposing solutions.

TEACHING EXPERIENCE

SI579: Building Interactive Applications

Jan 2022 - May 2022 | University of Michigan

Instructor: Ben Mullins

SI106: Programs, Information and People

Sep 2021 – Dec 2021 | University of Michigan

Instructor: Steve Oney

SI579: Building Interactive Applications

Jan 2021 – May 2021 | University of Michigan

Instructor: Steve Oney

SKILLS

- **Programming skills:** React, JavaScript, TypeScript, HTML5, CSS3, Git, Material-UI, Node.js ES6 and agile development. Strong in front-end development with good understanding of object-oriented programming.
- **UX Design skills:** Figma, Adobe Creative Suite, Miro, and Sketch for prototyping
- **Research skills:** Usability testing, Prototyping, Qualitative research methods and quantitative research methods.